

Cedric de Wijs

Senior Embedded Linux Software Developer

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Summary

Senior Embedded Linux & C++ Engineer with 20+ years of experience developing industrial and medical embedded systems. Specialized in embedded Linux, C/C++, real-time control systems, CI/CD automation, and hardware/software integration for high-reliability environments. Experienced in leading architecture decisions, mentoring engineers, and delivering robust systems for industrial automation and medical research applications.

Education

Hogeschool Inholland Haarlem

Bachelor of Electrical Engineering
Jul, 1999 - Jul, 2003

Core Skills

C++, Qt, Qwt development, Embedded Linux development, Gitlab CI/CD build servers and pipelines, Git, Gitlab and Github version control, Vertical farming equipment, CANopen debugging, Experience with distributed industrial control systems and fieldbus communication architectures, Yocto

Additional Skills

Embedded C development, Control systems engineering, EMC testing, PCBA design using Altium and KiCAD, Code reviews, Quality assurance systems, Kernel development, Device drivers, Cross-compilation, Networking, GDB, Profiling/debugging tools, Memory optimization, ARM architectures, Embedded filesystems, Secure update mechanisms

Soft Skills

Technical leadership and mentoring, Ability to explain complex concepts in simple terms, Ability to work independently and take a leading role,

Work Experience

Airlux Technologies

Mar, 2022 - Present

Senior Embedded Hardware Engineer

- Designed and delivered embedded control systems for climate-controlled environments in vertical farming, achieving $\pm 0.3^{\circ}\text{C}$ temperature and $\pm 3\%$ humidity stability
- Ported embedded firmware to Qt/C++ based simulation environments for automated testing and fast debug cycles.
- Ported embedded pic18 code to a 64-bit Linux environment, enabling automated unit testing and end-to-end testing on a build server.
- Established Github based CI/CD pipelines for automatic unit testing and firmware releases with versioning based on git tags and branch names.
- Performed code reviews, safeguarding software quality.
- Mentored colleagues on software architecture, unit testing frameworks, git workflows, and CI/CD pipelines.
- Analyzed and resolved complex software defects.
- Collaborated with international software, and hardware teams.
- Implemented PCBA improvements based on EMC testing, significantly improving signal integrity and consequently system accuracy.
- Environment: C, Unity, Pic18, C++, Qt, Github CICD, Linux, Git, Altium, KiCAD, FreeCAD, RS485, I2C, Modbus, Valgrind
- Quality Assurance processes: Test-driven development, Static analysis, Code coverage, Agile/Scrum, Release engineering, Branching strategies, Documentation standards, Architecture review, Code review

Leiden University Medical Center (LUMC)

Mar, 2013 - Mar, 2022

Senior Instrument Maker / Embedded Software Engineer

- Created a C++/Qt program for Circadian rhythm research. This program controls lights via a DALI bus, enabling researchers to create custom day/night patterns.
- Developed embedded Linux systems for remote research setups.
- Developed precision electronic systems for medical and research applications, for instance a low-noise control system achieving $\pm 0.1^{\circ}\text{C}$ stability for sensitive biological experiments.
- Created embedded software according to IEC 62304 medical software standards and MDR compliance requirements.
- Modified medical hardware, maintaining MDR compliance requirements.
- Introduced gitlab CI/CD pipelines and emulation environments for ATmega microcontrollers.
- Performed root-cause analysis and resolved complex hardware/software integration defects
- Collaborated closely with researchers, clinicians, and the mechanical team to translate requirements into robust solutions
- Provided technical guidance and mentoring to colleagues and trainees.

Ability to divide up complex systems into individual components

Languages

Dutch (native speaker), English (fluent), German (basic), French (basic)

- Environment: C, Unity, AtMEGA, emulation, C++, Qt, Gitlab CI/CD, Linux, Git, Altium, KiCAD, FreeCAD, DALI, I2C, RS232, Labview, IEC 62304

Quant Consultancy Electronics

Mar, 2008 - Mar, 2013

Hardware & Software Engineer

- Designed embedded software in C and assembly for reliable real-time systems
- Ported legacy code to more modern platforms (PIC16/18/24), reducing production costs.
- Conducted EMC testing and redesigns to improve compliance and reduce certification risk.
- Designed and delivered PCBA's for high volume production.
- Environment: C, Assembly, Pic16, Pic18, Pic24, Linux, Git, Altium, KiCAD, I2C, RS232

Beckhoff New Automation Technology

Mar, 2004 - Mar, 2008

Support Engineer

- Support customers using Beckhoff control systems based on field buses like CanOpen, Profibus, Ethercat.
- Resolved problems in complicated configurations and big PLC programs by reproducing them in minimal configurations, reporting them to the software team, and deliver the solutions back to the customer.
- Established a tracking system for support requests and solutions, virtually eliminating response times for common questions.
- Created documentation change tracking processes, reducing the time before I discovered new features of the supported hardware and software.
- Reviewed technical proposals for incompatible system combinations, reducing return and exchange costs
- Environment: TwinCAT, IEC 61131-3, Ethercat, CANopen, Profibus, Real-time ethernet, Lightbus

Selected Embedded Linux Project

High-Performance Multi-Channel DVR System

- Designed and developed a Linux-based Digital Video Recorder capable of recording 22 DVB-T channels simultaneously.
- Patched Linux kernel drivers to enable the system to suspend and resume reliably, when an USB receiver that's in use is disconnected from the system.
- Ported the system from Arch Linux on x86 hardware to Debian running on an Allwinner A20 ARM SoC, reducing power consumption by over 90%
- Documented the development process on the Arch Linux and Debian wiki platforms
- Environment: VDR, Debian, Arch Linux, Allwinner A20, Digipenne

Volunteer Experience

- Volunteering at open source conferences including FOSDEM (Belgium) and FrOSCon (Germany)
- Volunteering at Linux repair cafés by migrating systems from Windows to Linux while maintaining the ability to run needed Windows programs.
- Volunteering at LaptopRevive.nl by installing Linux on donated laptops for students that cannot afford them.
- Founder of the soccer club "Terrible football Haarlem"